

Engineering quality audit

Executed baseline

Check	Result
Clean install	Root/server/mobile/infrastructure lockfiles installed; mobile took ~2 minutes
Unit tests	PASS: 87 files, 729 tests
Web staging build	PASS; oversized bundle warning
API TypeScript build	PASS
CDK TypeScript build/synth	PASS; 78 feature flags unconfigured
Mobile type-check	FAIL: getReactNativePersistence is not exported from firebase/auth
Lint	FAIL: 1,004 problems (871 errors, 133 warnings) in root run
Integration tests	FAIL/BLOCKED: 8 files failed, 52 skipped; Postgres unavailable because Docker daemon was stopped
E2E	Missing/broken: no specs and configured package absent (independent review)

Scale model

Peak	Expected limiting factors	Evidence/confidence
100 concurrent	Current staging max two 0.5-vCPU tasks; DB/cache likely adequate after query fix	Probable; no cloud load test
1,000 concurrent	Rate-limit identity/proxy ambiguity, single Redis, external API latency, Fargate scale lag	High risk
10,000 concurrent	Max 10 prod tasks, DB connection/query capacity, in-process jobs, WS fanout, provider quotas	Not demonstrated
100K-1M MAU	Architecture can evolve, but current fixed services, event volume, support/privacy and vendor quotas are unproven	Missing evidence

The attached simulation claims 375 req/s after a query fix. Raw scratchpad/sim artifacts are absent, so the result is accepted only as a reported experiment, not reproduced evidence.

Findings

ID	Domain	Title	Severity	Confidence	Evidence	User/business impact	Recommended correction	Effort	Owner	Release blocker	Status
ENG-001	Mobile	Native code does not type-check	High	Confirmed	executed npm run typecheck	No reliable store build	Correct Firebase RN import/version and test build	Low/Medium	Mobile	yes	Open
ENG-002	Release	No tracked CI/CD quality gates	High	Confirmed	workflows explicitly ignored	Unsafe/repeatable releases impossible	Tracked protected pipeline	Medium	Release	yes	Open
ENG-003	QA	E2E harness empty/broken	High	Confirmed	no specs; missing config package	Critical journeys untested	Add Playwright/native E2E	Medium	QA	yes	Open
ENG-004	Quality	Lint gate fails at scale	Medium	Confirmed	871 errors	Defects/noise hide regressions	Scope generated files; burn down tracked errors	Medium	Leads	before beta	Open

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ENG-005	Test	Integration suite not reproducible without local services	Medium	Confirmed blocker	Docker unavailable, ECONNR EFUSED	DB integrity unverified	Hermetic CI service containers	Medium	QA/DevOps	yes	Open
ENG-006	Jobs	Timers run once per ECS task	Medium	Confirmed	API startup intervals	Duplicate reads/writes/cost	EventBridge singleton + lease	Medium	Backend	before 1K	Open
ENG-007	Reliability	Redis is one non-TLS non-HA node	High	Confirmed	CfnCacheCluster	Rate limit/cache/fan out outage	Replication group, TLS, auth, failover	Medium	SRE	before real data	Open
ENG-008	Observability	No tracing, SLOs, synthetic journeys or restore drill	Medium	Confirmed/missing evidence	repo/laC	Slow incident diagnosis/recovery	OTel/correlation/SLO/restore exercises	High	SRE	before 1K	Open
ENG-009	Dependencies	Production trees contain critical/high advisories	High	Confirmed	npm audit --omit=dev	Supply-chain/runtime exposure	Reachability triage and upgrades	Medium	Security/leads	yes	Open
ENG-010	Maintainability	Legacy/prototype/provider branches coexist	Low	Confirmed	Supabase/redesign/mock mappings	Hidden coupling/ambiguity	Archive and document source of truth	Medium	CTO	no	Open

Positive evidence

- * Payment and booking tests cover idempotency, price reconciliation, refunds and concurrency-oriented constraints.
- * PostgreSQL calls are predominantly parameterized; pool/transaction cleanup exists.
- * Deployment circuit breaker, healthy-percent controls, auto-scaling and alarms are present.
- * Public marketplace query caching and rating denormalization target a measured bottleneck.